

MOBILE APPLICATION SYSTEM DESIGN

(Week 1)

Nurul Amera Muhamad Nazmi

CONTENT

1. Course introduction
2. Introduction to Mobile Application
3. Operating system for Android and iOS
4. Mobile application market trends and statistic
5. Mobile application research and design
6. Class Discussion : What is your dream app?
7. Project Management and Tools

COURSE INTRODUCTION - LEARNING OUTCOMES

- Demonstrate ability to apply the mobile application architecture.
- Evaluate navigation schemes in mobile application.
- Analyze related programming languages to develop basic features in mobile application.
- Display an awareness for continuous learning and analysis of mobile application technology.

CONTINUOUS ASSESSMENT

- Assignment 1 (40%)
- Assignment 2 (30%)

Final Assessment

- Final project (30%)

Course Introduction – Expectation

- Punctual
- Alert to course announcements and assessment instructions
- No late submission of assessment tasks without a valid reason

Course Introduction – Lecturer Contact Information

- Lecturer: Nurul Amera
- Email: nurulamera.nazmi@xmu.edu.my
- Consultation Hours (Appointment basis)
 - Friday: 12 pm – 3 pm
 - Thursday: 10am – 11am

(Please email or Team Chat me before coming for consultation)

INTRODUCTION TO MOBILE APPLICATION SYSTEM DESIGN

- What is a mobile application?
 - A mobile application, most commonly called an app, is software designed to run on a mobile device, such as a smartphone or tablet computer.
 - Mobile applications frequently provide users similar services to those accessed on PCs. Apps are generally small, individual software units with limited functions.



Introduction to Mobile Application

- Three main types of mobile apps:
 - Native
 - Web
 - Hybrid

Introduction to mobile application

Native Mobile App

- ❑ App that is developed for one particular operating system
- ❑ Native mobile apps are designed to be “native” to one platform, whether it’s Apple’s iOS, Google’s Android, or others.
- ❑ The native platform can be advantageous because it optimizes the user experience.
- ❑ Because it was developed specifically for the platform, it can operate more quickly and intuitively.



Introduction to mobile application

NATIVE APPS

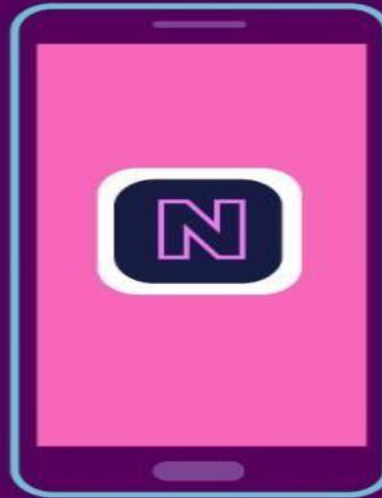
TECHNOLOGY USED

Java, Kotlin, Python, Swift, Objective C, etc.



PROS

- 1 Faster, better performance
- 2 Native UI
- 3 Can access device features



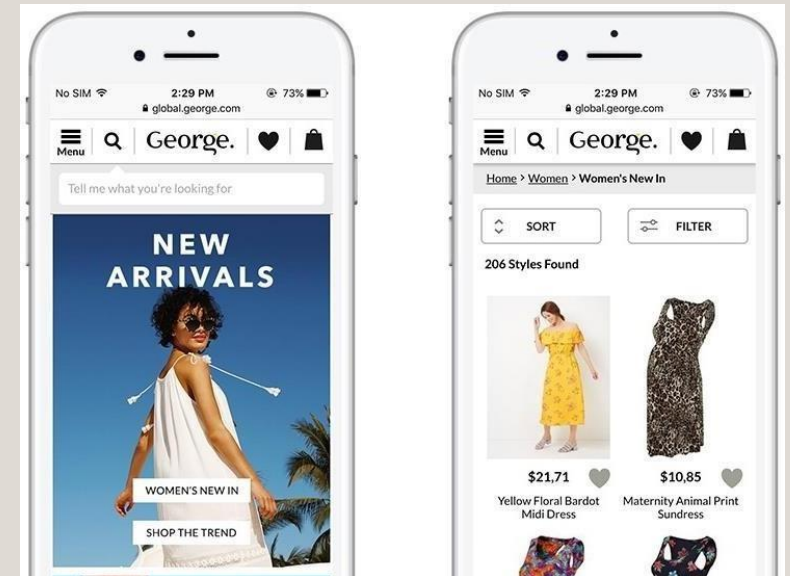
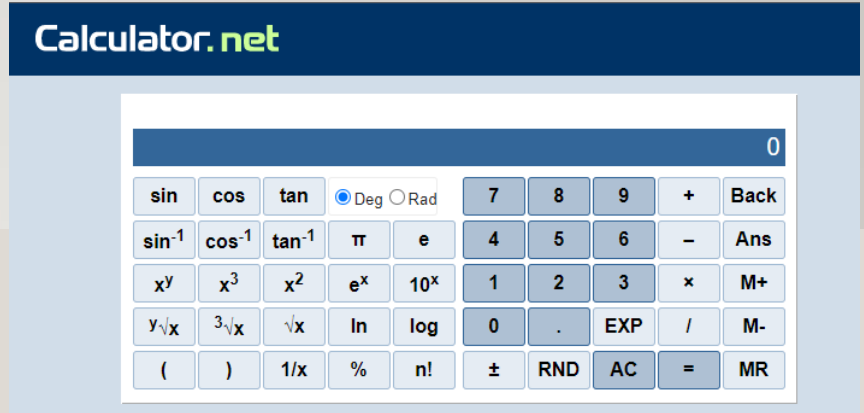
CONS

- 1 Higher cost to maintain
- 2 Takes up space in the device
- 3 Updates must be downloaded

INTRODUCTION TO MOBILE APPLICATION

Web Apps

- ❑ An application program that is stored on a remote server and delivered over the internet through a browser interface.
- ❑ Commonly used web applications can include webmail, online calculators or e-commerce shops.
- ❑ Users don't need to install the app and can access the app through various platforms such as a desktop, laptop or mobile via browser.
- ❑ However, a web app operates relatively at a slower speed than the application hosted on a local server
- ❑ As web applications are not native, they cannot sometimes effectively collaborate with all the hardware and operating systems of the specific devices you use.



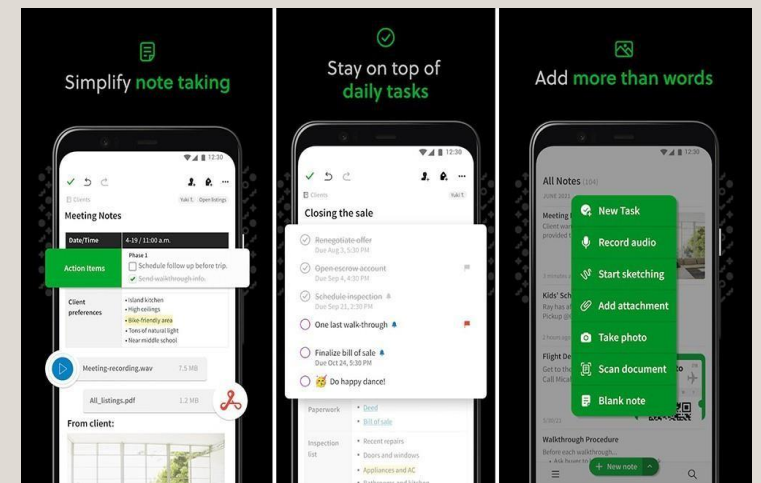
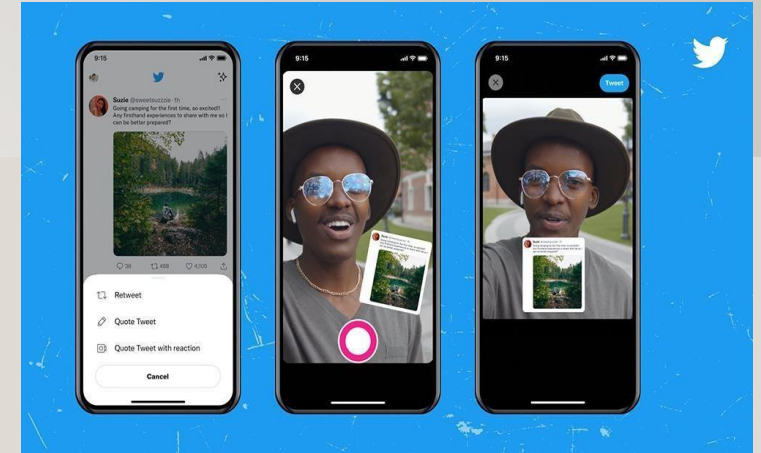
INTRODUCTION TO MOBILE APPLICATION



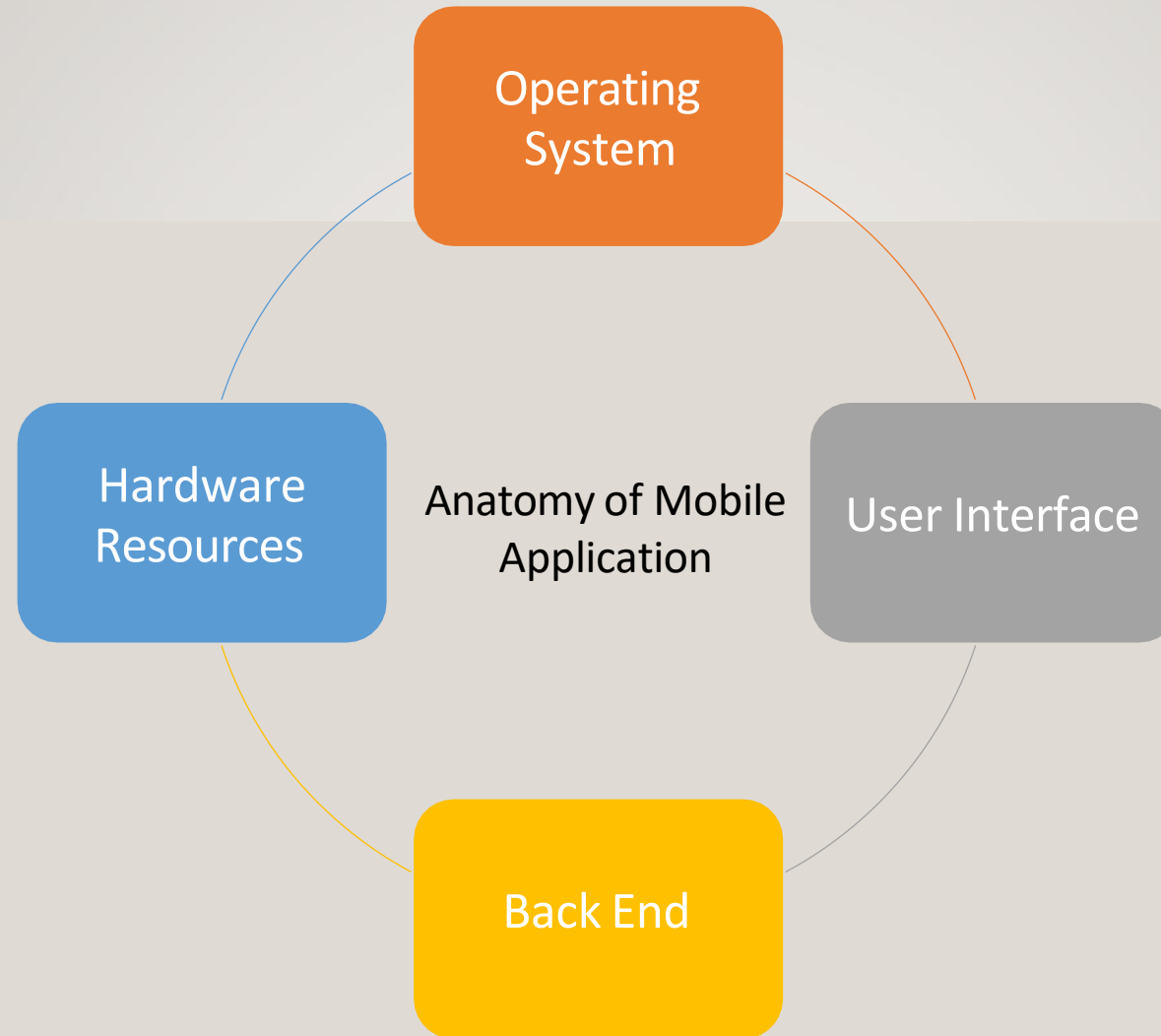
INTRODUCTION TO MOBILE APPLICATION

Hybrid Apps

- ❑ A hybrid mobile app combines the elements of a native app and a web app.
- ❑ The application's core is written using web technologies (HTML, CSS, and JavaScript), which are then encapsulated within a native application.
- ❑ It is run from within a native application and its own embedded browser, essentially invisible to the user.
- ❑ Through the use of plugins, these applications can have full access to the mobile device's features.
- ❑ Users can download the app from an app store and install it locally



ANATOMY OF MOBILE APPLICATION



ANATOMY OF MOBILE APPLICATION

Operating system

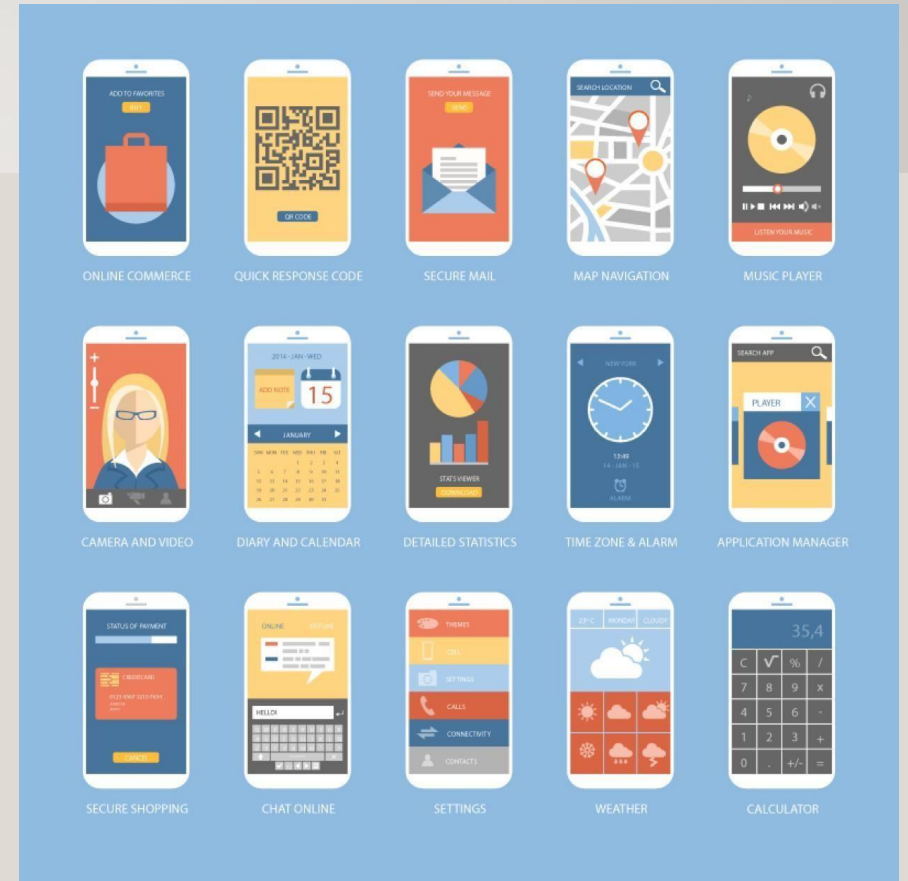
- Majority of smart phones run either Android (developed by Google) or iOS (Apple's mobile OS).
- The operating system controls the whole phone, in particular providing access to hardware functions such as the GPS, data network, motion sensors and storage.
- It also coordinates things like notifications (more on those later).



ANATOMY OF MOBILE APPLICATION

User Interface

- The user interface or UI is a combination of what you see when you load and run your app and the way you interact with it.
- The different elements of the UI combine to provide you with a good user experience (or UX).



ANATOMY OF MOBILE APPLICATION

Back End

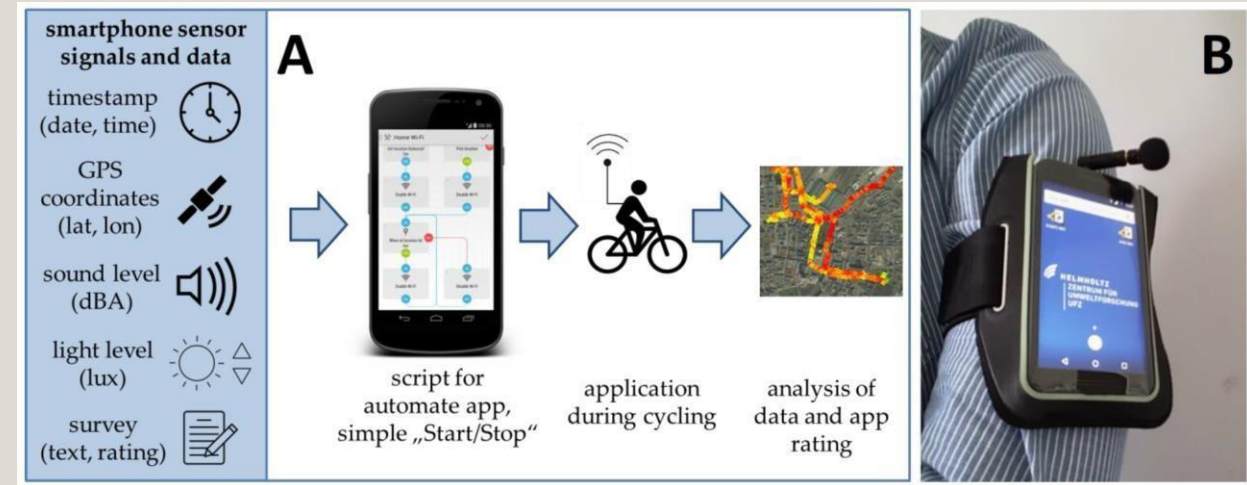
- A back-end service provides the functionalities that the app needs to store and manage data, authenticate users, process and analyze data, and communicate with other systems or services.



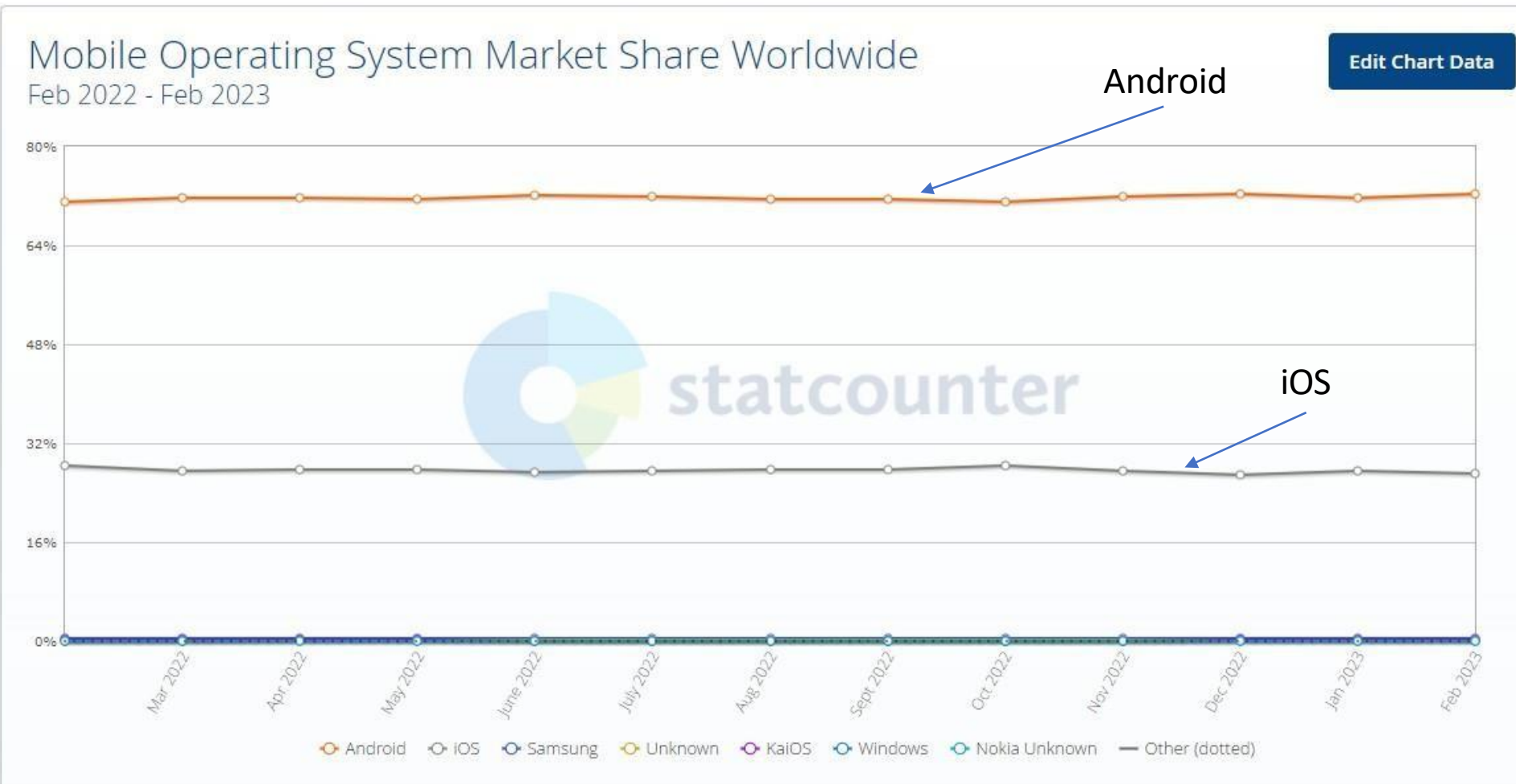
ANATOMY OF MOBILE APPLICATION

Hardware Resources

- On most modern phones there are a huge array of sensors and services that the app could access.
- These hardware include GPS for giving your position, accelerometers to track the movement of your phone, the speaker or other audio outputs, the microphone and the camera(s)



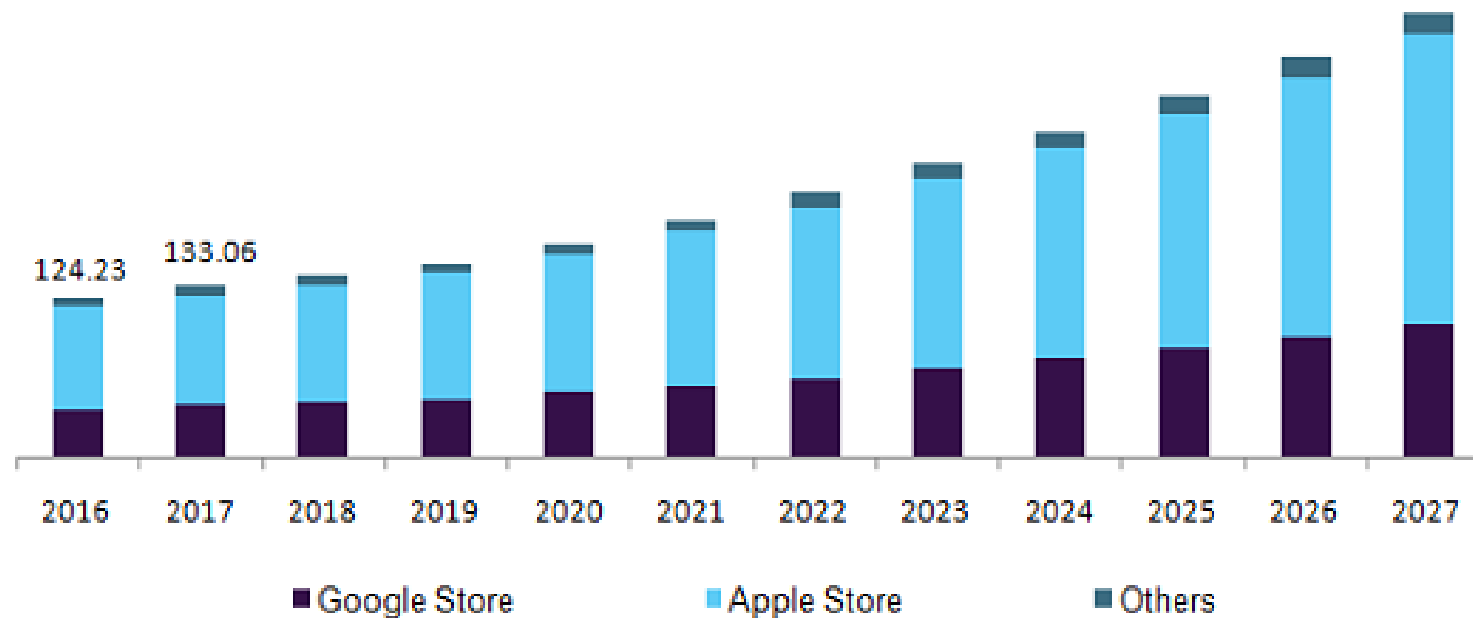
MOBILE APPLICATION MARKET TRENDS AND STATISTIC



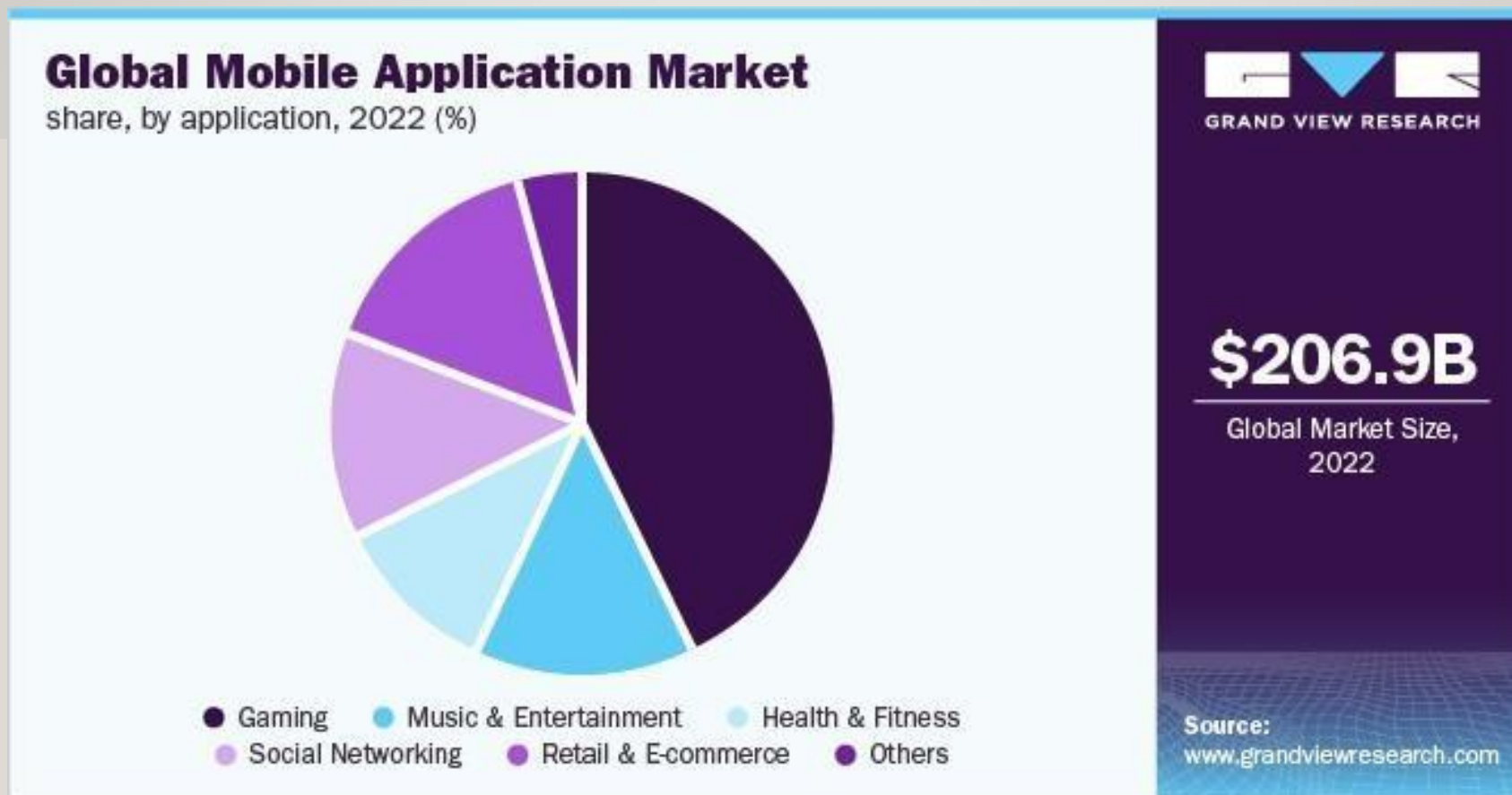
Source: <https://gs.statcounter.com/os-market-share/mobile/worldwide>

MOBILE APPLICATION MARKET TRENDS AND STATISTIC

Global mobile application market size, by store type, 2016 - 2027 (USD Billion)



MOBILE APPLICATION MARKET TRENDS AND STATISTIC



MOBILE APPLICATION MARKET TRENDS AND STATISTIC

Top Mobile Games by Worldwide Downloads for August 2022

SensorTower

OVERALL DOWNLOADS

- | | | | |
|----|---|---|---|
| 1 |  | Subway Surfers
MINICLIP | △ |
| 2 |  | Stumble Guys
KITKA GAMES | ▽ |
| 3 |  | Garena Free Fire
GARENA | — |
| 4 |  | GT Car Stunt Master 3D
ZEGO GLOBAL | ★ |
| 5 |  | Candy Crush Saga
KING | △ |
| 6 |  | Ludo King
GAMETION | ▽ |
| 7 |  | Roblox
ROBLOX | ▽ |
| 8 |  | Save the Doge
WONDER GROUP | ★ |
| 9 |  | School Party Craft
CANDY ROOM GAMES | ★ |
| 10 |  | Snow Race
CYBERAGENT | ★ |

APP STORE DOWNLOADS

- | | | | |
|----|---|---|---|
| 1 |  | Subway Surfers
MINICLIP | △ |
| 2 |  | Stumble Guys
KITKA GAMES | ▽ |
| 3 |  | Survivor!.io
HABBY | ★ |
| 4 |  | Roblox
ROBLOX | ▽ |
| 5 |  | Fishdom
PLAYRIX | △ |
| 6 |  | Tower of Fantasy
PERFECT WORLD | ★ |
| 7 |  | PUBG Mobile
TENCENT | ▽ |
| 8 |  | Honor of Kings
TENCENT | ▽ |
| 9 |  | Call of Duty: Mobile
ACTIVISION | △ |
| 10 |  | Candy Crush Saga
KING | △ |

GOOGLE PLAY DOWNLOADS

- | | | | |
|----|---|---|---|
| 1 |  | Subway Surfers
MINICLIP | △ |
| 2 |  | Stumble Guys
KITKA GAMES | — |
| 3 |  | Garena Free Fire
GARENA | ▽ |
| 4 |  | GT Car Stunt Master 3D
ZEGO GLOBAL | ★ |
| 5 |  | Ludo King
GAMETION | ▽ |
| 6 |  | Candy Crush Saga
KING | △ |
| 7 |  | Save the Doge
WONDER GROUP | ★ |
| 8 |  | School Party Craft
CANDY ROOM GAMES | ★ |
| 9 |  | Snow Race
CYBERAGENT | ★ |
| 10 |  | Roblox
ROBLOX | ▽ |

NOTE: DOES NOT INCLUDE DOWNLOADS FROM THIRD-PARTY ANDROID STORES IN CHINA OR OTHER REGIONS △ = RANK UP OVER LAST MONTH ▽ = RANK DOWN OVER LAST MONTH ★ = NEW TO TOP CHART

MOBILE APPLICATION RESEARCH AND DESIGN

- Market research is the first step of every mobile app development project.
- Mobile app market research helps validate your app idea and ensures your app is addressing existing market needs.



MOBILE APPLICATION RESEARCH AND DESIGN

User Experience (UX) and UX Research

- User Experience refers **to the feeling users experience when using a product, application, system, or service.**
- UX (user experience) research is the systematic investigation of users and their requirements, in order to add context and insight into the process of designing the user experience.
- UX research employs a variety of techniques, tools, and methodologies to reach conclusions, determine facts, and uncover problems, thereby revealing valuable information which can be fed into the design process.



MOBILE APPLICATION RESEARCH AND DESIGN

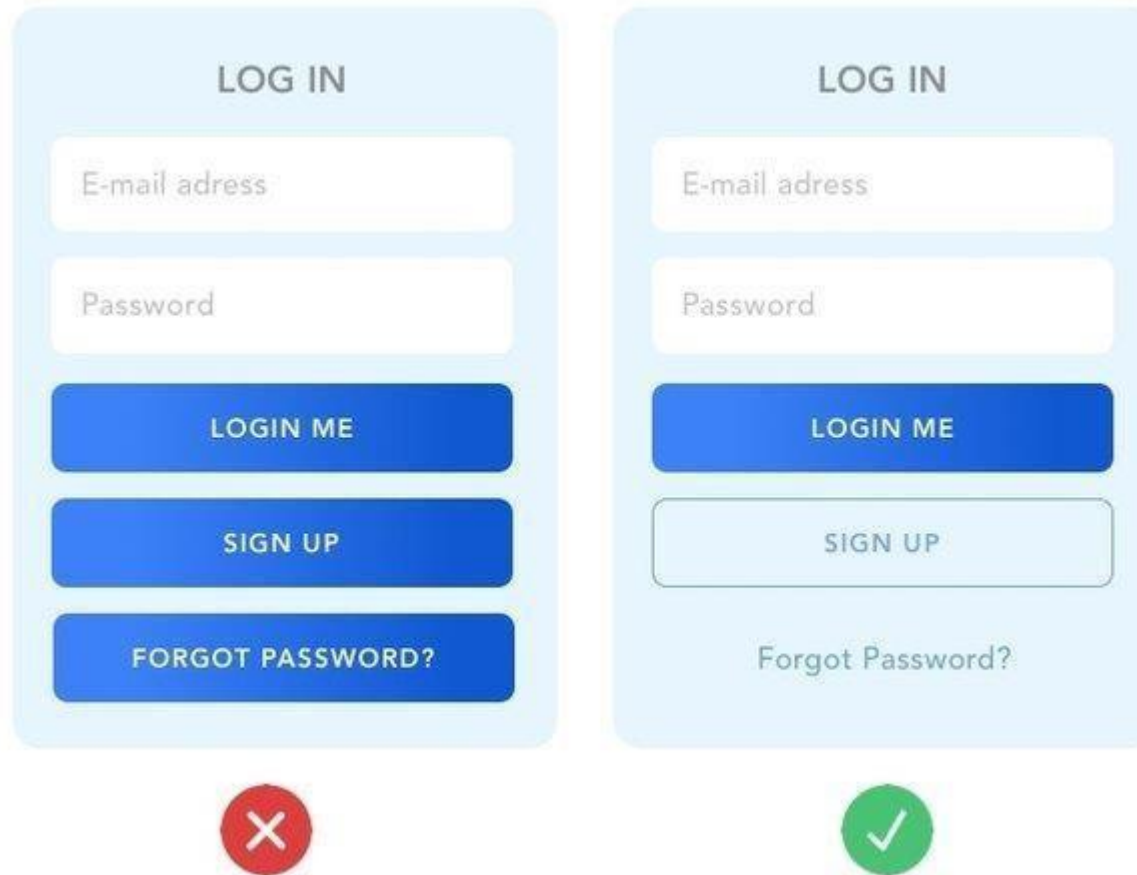
Why UX research is important



MOBILE APPLICATION RESEARCH AND DESIGN



Example of good and bad UX design



The image compares two login form designs side-by-side. Both forms are titled 'LOG IN' and contain two input fields: 'E-mail adress' and 'Password'. Below the inputs are three options: 'LOGIN ME', 'SIGN UP', and 'FORGOT PASSWORD?'. The left form (bad UX) has all three options as solid blue buttons. The right form (good UX) has 'LOGIN ME' as a solid blue button, while 'SIGN UP' and 'FORGOT PASSWORD?' are light blue buttons with thin borders. Below each form is a circular icon: a red circle with a white 'X' for the bad design and a green circle with a white checkmark for the good design.

LOG IN

E-mail adress

Password

LOGIN ME

SIGN UP

FORGOT PASSWORD?

LOG IN

E-mail adress

Password

LOGIN ME

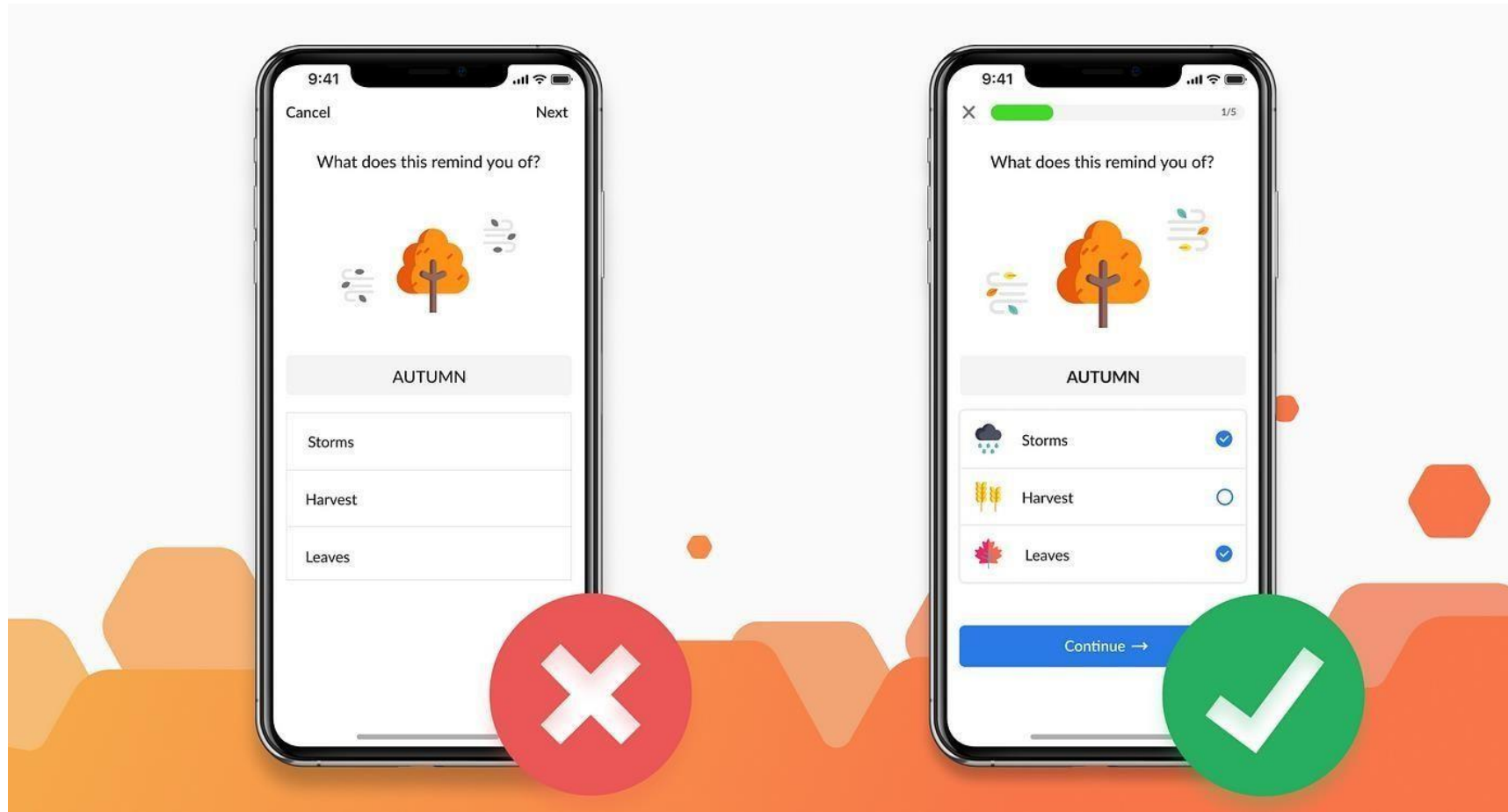
SIGN UP

Forgot Password?

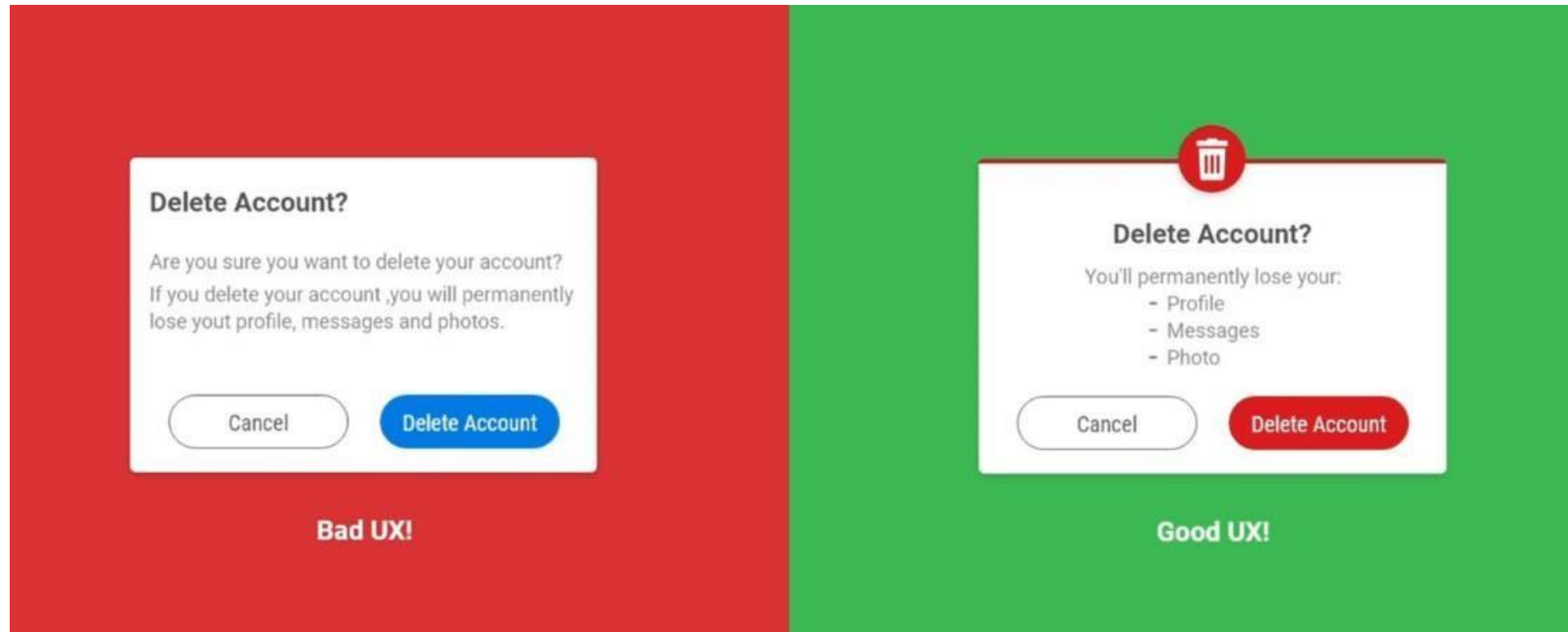
✗

✓

Example of good and bad UX design



Example of good and bad UX design



User Research for Mobile Applications.

Qualitative User Research Methods for Mobile Apps

1. Lab Studies
2. Journal/Diary Studies
3. Observational/Field Studies

Quantitative User Research Methods for Mobile Apps

1. Online Surveys (including Behavioral Tracking)
2. Behavioural Analysis
3. Experience Sampling Method (ESM)

MOBILE APPLICATION RESEARCH AND DESIGN

Qualitative User Research Methods : Lab Studies

- Lab studies are designed to test whether the user can actually use the product.
- A lab study is conducted using a mobile device in an environment where the user can be observed by the research team.
- This will often involve the researcher remaining in a different room observing remotely while the user goes about their tasks.



Qualitative User Research Methods: Observational/Field Studies

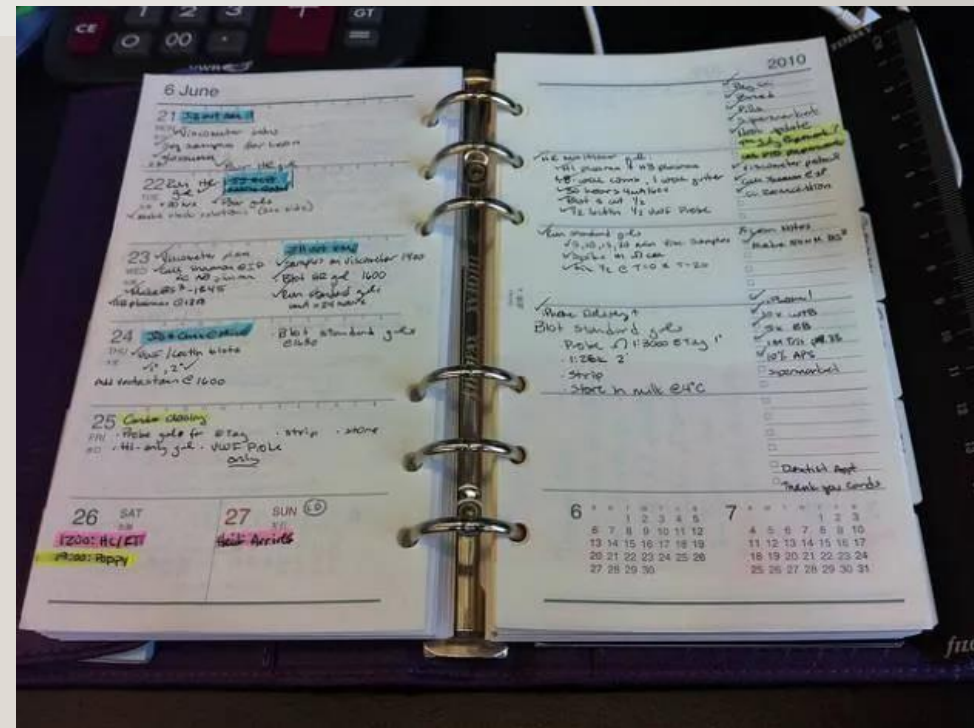
- Researchers meet with test participants in their natural environment (where they would most likely encounter the product), observe how users interact with a product, and ask related questions.



MOBILE APPLICATION RESEARCH AND DESIGN

Qualitative User Research Methods: Journal/Diary Studies

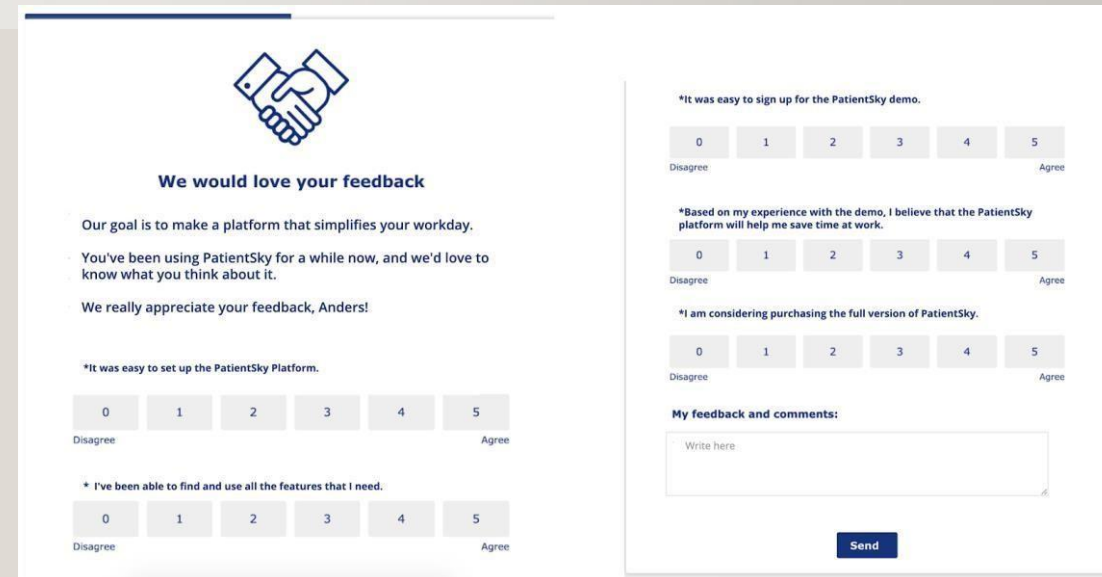
- Test participants are given a mechanism to record and describe aspects of their lives relevant to a product.
- The diary can be physical or online. Typically, test participants document their activities over a prolonged period.



MOBILE APPLICATION RESEARCH AND DESIGN

Quantitative User Research Methods : Online Surveys (including Behavioural Tracking)

- Online surveys will involve using a questionnaire combined with automated data collection reflecting the user's actual behaviour.
- They are designed to answer the questions; “What do users think of our product? What do they perceive the value to be?”
- Online surveys are incredibly cost-effective and can create large amounts of data on a geographically diverse basis which, in turn, leads to statistically significant sample sizes and results.



The image shows a feedback survey form for PatientSky. It features a handshake icon at the top, followed by the heading "We would love your feedback". The form contains three Likert scale questions, each with a 0 to 5 rating scale. The first question is "It was easy to sign up for the PatientSky demo." with a rating of 5. The second question is "Based on my experience with the demo, I believe that the PatientSky platform will help me save time at work." with a rating of 5. The third question is "I am considering purchasing the full version of PatientSky." with a rating of 5. Below the scales is a text box for "My feedback and comments:" and a "Send" button.

We would love your feedback

Our goal is to make a platform that simplifies your workday.

You've been using PatientSky for a while now, and we'd love to know what you think about it.

We really appreciate your feedback, Anders!

*It was easy to set up the PatientSky Platform.

0 1 2 3 4 5
Disagree Agree

*I've been able to find and use all the features that I need.

0 1 2 3 4 5
Disagree Agree

*It was easy to sign up for the PatientSky demo.

0 1 2 3 4 5
Disagree Agree

*Based on my experience with the demo, I believe that the PatientSky platform will help me save time at work.

0 1 2 3 4 5
Disagree Agree

*I am considering purchasing the full version of PatientSky.

0 1 2 3 4 5
Disagree Agree

My feedback and comments:

Write here

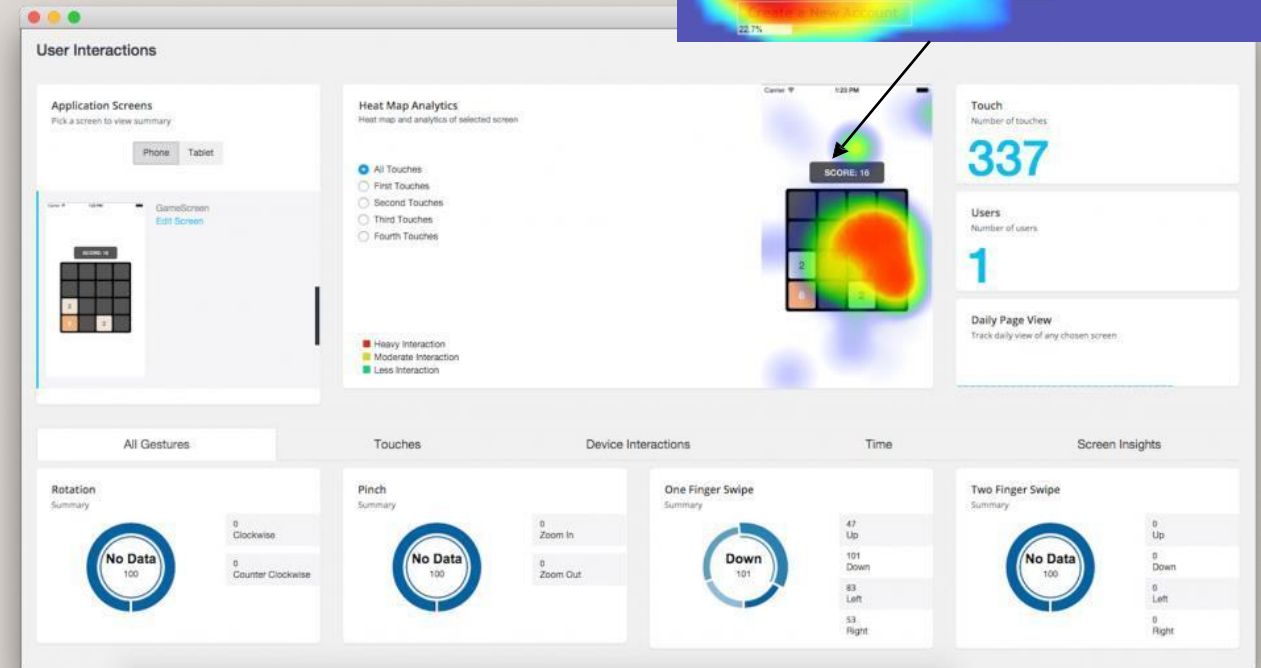
Send

MOBILE APPLICATION RESEARCH AND DESIGN

Quantitative User Research Methods: Behavioural Analysis

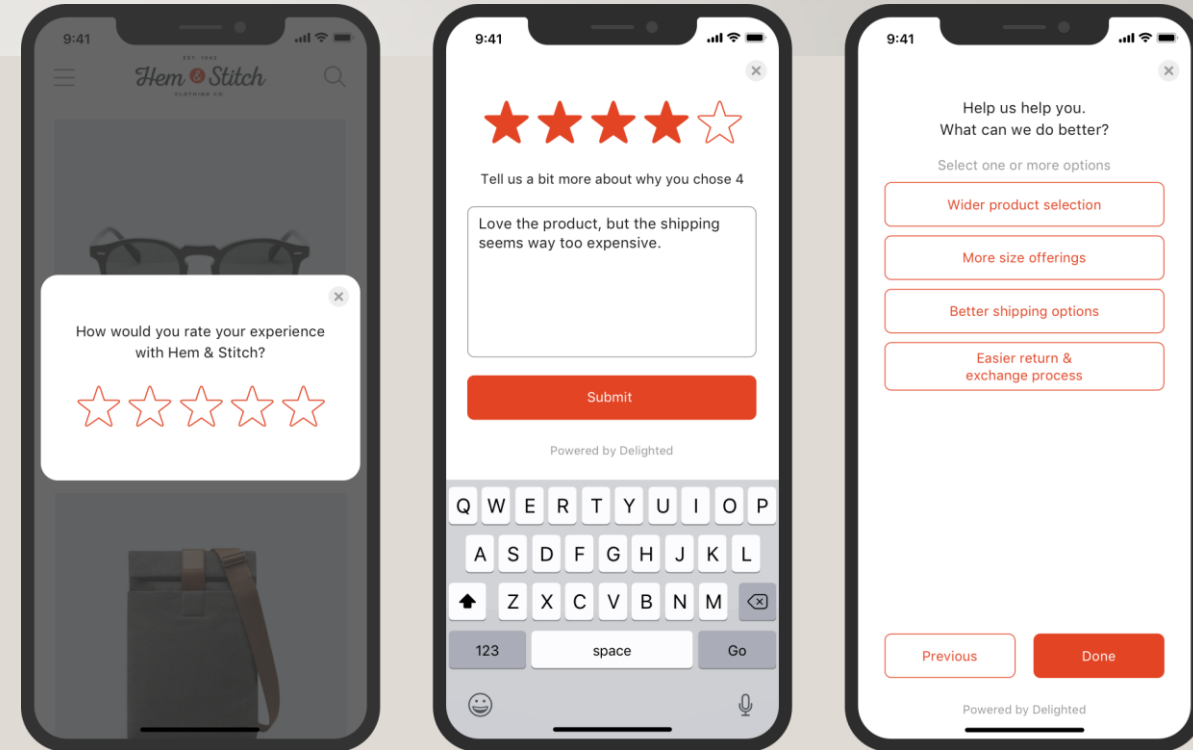
- Behavioral analysis works to determine the answer to the “How do people use our app?” question.
- It is conducted via installed software on the users’ devices that tracks how people navigate the app and how they conduct the tasks on your app.

Interaction heatmap



Quantitative User Research Methods: Experience Sampling Method (ESM)

- This method is beneficial in mobile user research because it consists of sending automated messages to the study participants at the point they carry out an action to ask about their experience.
- It helps answer two main questions; “How does the user use our app?” and “What do they think about that?”



CLASS DISCUSSION :WHAT IS YOUR DREAM APP?

- What is the current issue and problem that need to solve?
- Is it possible to solve with using mobile application?
- Write on a note and state the current issue, your idea and what is possible solution you have?
- All student to share their opinion and perspective.

[-padlet-](#)

MOBILE APP DEVELOPMENT FRAMEWORK

Native App Framework

- **Flutter**

- Flutter is a free and open-source mobile UI framework created by Google and released in May 2017
- Flutter is used by developers and organizations worldwide and is free and open source.
- To develop with Flutter, you will use a programming language called **Dart**.



<https://flutter.dev/>

Official Website

MOBILE APP DEVELOPMENT FRAMEWORK

Why Flutter?

- **One codebase for all platforms**

- Not necessarily writing code for Android and another codebase for iOS devices.
- Flutter's code reusability allows you to write just one codebase and use it for mobile Android and iOS, web, desktop, and more.
- This significantly cuts development time, removes costs, and enables you to launch your app much faster.



MOBILE APP DEVELOPMENT FRAMEWORK

Why Flutter?

- **Rich libraries**

- It has a rich graphics library, which is fast and mature enough to create stunning user interfaces.

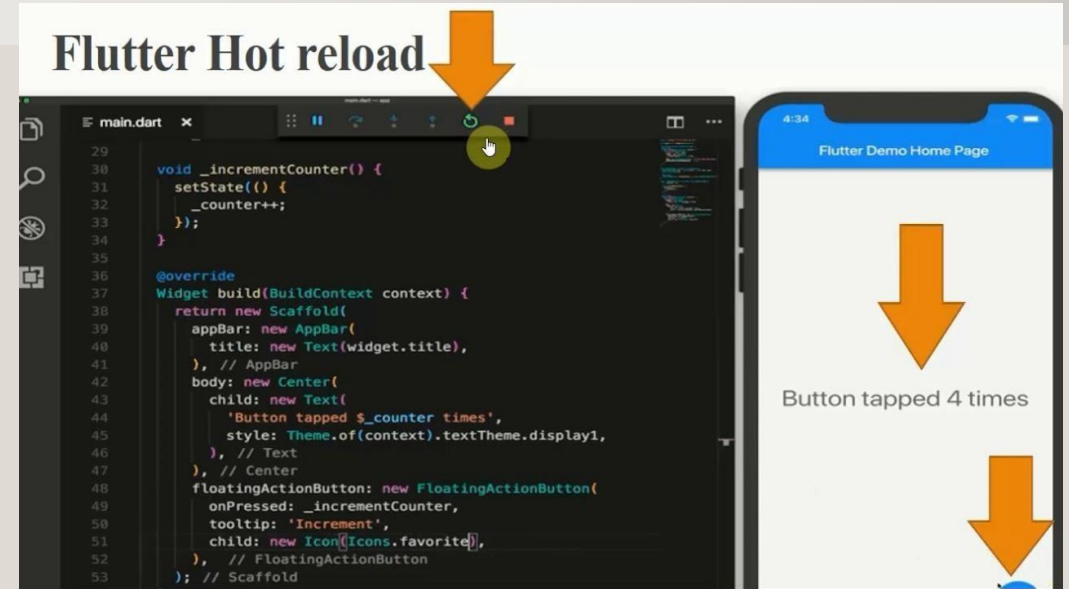


MOBILE APP DEVELOPMENT FRAMEWORK

Why Flutter?

- **Fast testing with hot reload**

- The hot reload feature makes the app development much quicker.
- With Flutter, reloading the app to see every change you make in the code is unnecessary.
- You can easily make changes in your app in real-time, allowing you to experiment with the code and fix bugs on the go.

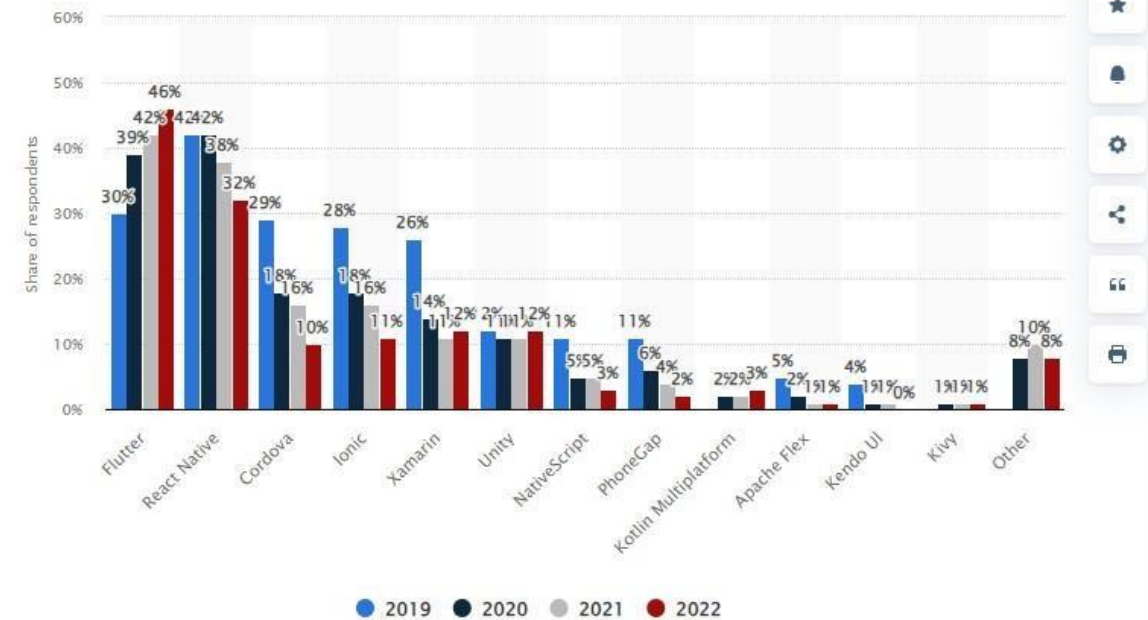


Why Flutter?

- **Popularity**

- One of the most popular cross-platform mobile frameworks used by developers worldwide.
- The popularity is increasing from year to year.
- This means Flutter has a very large community group where we can easily get support to troubleshoot some development issues.

Cross-platform mobile frameworks used by software developers worldwide from 2019 to 2022



MOBILE APP DEVELOPMENT FRAMEWORK

Why Flutter?

- **Great Job Opportunities**

- A high demand skill in the current job market for a mobile app developer
- Flutter or React Native is commonly one of the preferable skills required
- Stand a better chance to get a high salary job with years of experience.



Mobile Application Developer (Flutter)
MiCare Sdn Bhd
Selangor
Developers/Programmers
(Information & Communication Technology)

- Work with one of leading TPA for medical insurance in Malaysia
- Great career development
- Welcome self motivated and goal oriented person

5d ago

Mobile Developer / Senior Mobile Developer (MYR 5,400 ...)
iFAST Capital Sdn Bhd

[Quick apply](#) [Save](#)

- Bachelor's Degree in Computer Science or related field.
- Experience in Android/iOS application development.
- Must be analytical, self-driven and proactive in keeping up with new technologies and programming languages.
- Must have excellent problem solving skills and love technical challenges.
- Must be able to take initiative to be productive and efficient.
- Fresh Graduates? As long as you are willing to learn, we will consider your application!

Extra points when we assess your applications:

- Ability to showcase apps developed previously.
- Knowledge and exposure in modern frameworks (Angular/Flutter).



Senior Mobile App Developer, Flutter (MY)
REDFIN TECHNOLOGIES PL
Kuala Lumpur
RM 12,000 – RM 18,000 per month
Developers/Programmers
(Information & Communication Technology)

- Great new working exposure
- Great career succession planning
- Attractive Salary & Benefits

15d ago



Senior Mobile Application Developer (Flutter, React Native) (ID: 603180)
Agensi Pekerjaan PERSOLKELLY Malaysia Sdn Bhd
Mid Valley City, Kuala Lumpur
RM 7,500 – RM 11,000 per month
Developers/Programmers
(Information & Communication Technology)

8d ago

Hardware Requirement of Flutter (Windows)

Requirement	Minimum	Recommended
x86_64 CPU Cores	4	8
Memory in GB	8	16
Display resolution in pixels	WXGA (1366 x 768)	FHD (1920 x 1080)
Free disk space in GB	11.0	60.0

Hardware Requirement of Flutter (MacOS)

Requirement	Minimum	Recommended
CPU Cores	4	8
Memory in GB	8	16
Display resolution in pixels	WXGA (1366 x 768)	FHD (1920 x 1080)
Free disk space in GB	44.0	70.0

SEE YOU NEXT
CLASS